

Name : _____

Teacher : _____

SYDNEY TECHNICAL HIGH SCHOOL

MATHEMATICS

Year11

PRELIMINARY ASSESSMENT

TASK 2

JULY 2003

Time allowed: 70 minutes

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
/8	/8	/8	/8	/8	/8	/9	/9	/66

Instructions:

- Show all necessary working in every question.
- Attempt all questions.
- All questions are not of equal value.
- Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- These questions are to be handed in with your answers.

QUESTION 1: (8 marks)

(1) a. Write down the exact value of $\tan 30^\circ$

(3) b. Factorise i. $x^3 - 27$

ii. $x^2 - xy - 3x + 3y$

(2) c. Solve $|2x+1| \leq 7$

(2) d. Solve $3x^2 - 14x - 5 = 0$

QUESTION 2: (8 marks)

- (2) a. Rationalise the denominator of

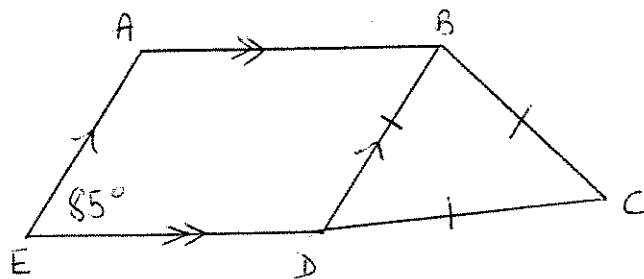
$$\frac{3}{\sqrt{7}+2}$$

- (3) b. Solve simultaneously

$$2x + 3y + 1 = 0$$

$$3x - y = 4$$

- (3) c. Find the value of $\angle ABC$, giving reasons.



Not to scale

QUESTION 3: (8 marks)

- (6) a) Write down the exact value of:

i. $\cos 225^\circ$

ii. $\sec(-30^\circ)$

iii. $\tan^2 390^\circ$

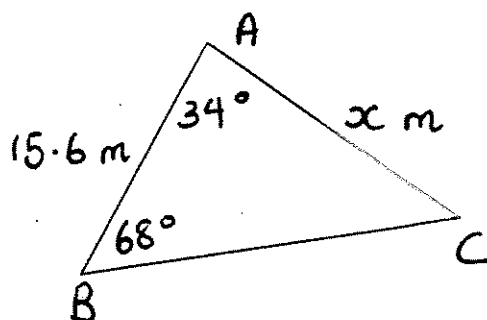
- (2) b) If $\cos \theta = p$ and $\sin \alpha = q$, find an expression for,

$$\cos(360^\circ - \theta) \cdot \sin(180^\circ + \alpha)$$

QUESTION 4:**(8 marks)**

(1) a. Solve $\sin(2x)^\circ = \cos(40 + x)^\circ$, where $0^\circ \leq x \leq 90^\circ$

(5) b.



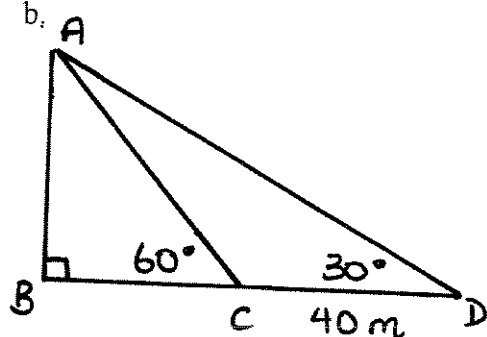
- i. Find x , correct to the nearest centimetre.
- ii. Calculate the area of $\triangle ABC$, correct to the nearest square metre.

- (2) c. The midpoint M of the interval AB is $(1, -3)$
Find the co-ordinates of B given A is $(7, 2)$

QUESTION 5:**(8 marks)**

- (3) a. If $\sin \alpha = m$ and α is obtuse, find an expression for $\cos \alpha$

(5) b.



- i. Write down the size of $\angle ACD$
- ii. Find the length of AC
- iii. Find the exact length of AB

QUESTION 6: (8 marks)

(2) a. Solve $\cos x = -\frac{1}{2}$ for $-180^\circ \leq x \leq 180^\circ$

(3) b. Solve $3\tan^2 x - 1 = 0$ for $0^\circ \leq x \leq 360^\circ$

(3) c. Solve $\sec 2x = 1$ for $0^\circ \leq x \leq 360^\circ$

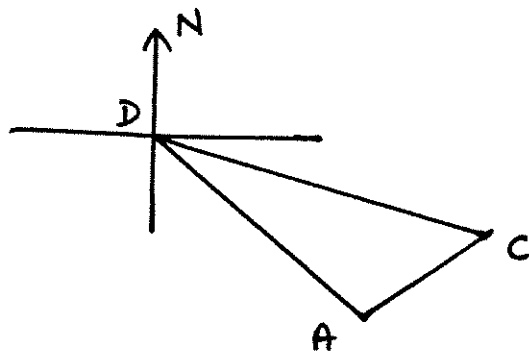
QUESTION 7: (9 marks)

(4) a. Prove that

$$\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} = 2 \operatorname{cosec} \theta$$

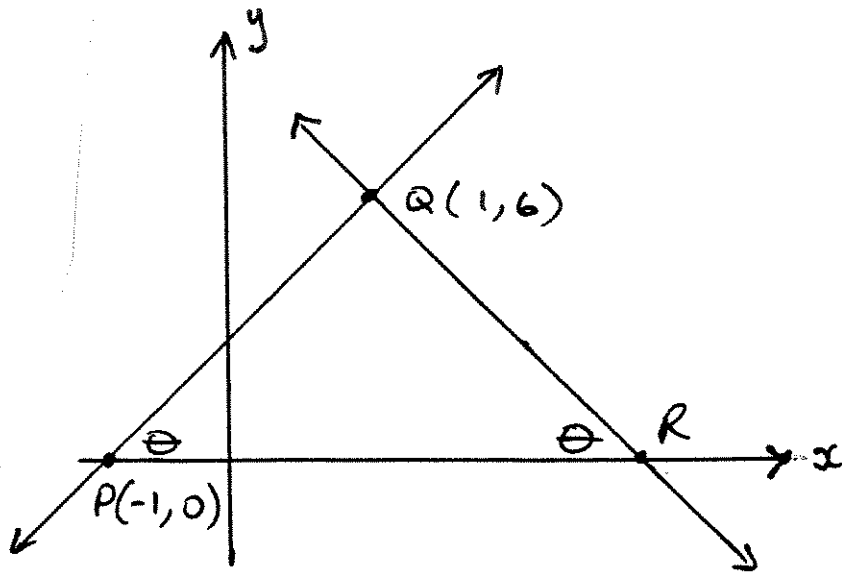
(5) b. Daffy walks on a bearing of 150° T, from D, for 12km before changing his course at

A to 040° T and continuing for 8km to C.



- i Copy the diagram and explain why $\angle DAC = 70^\circ$
- ii How far is Daffy from his starting point?
- iii What is the bearing of C from Daffy's starting point D?

QUESTION 8: (9 marks)



Triangle PQR is isosceles with
 $P(-1, 0)$, $Q(1, 6)$

- (1) a. Find the distance of PQ
- (2) b. Find the equation of the line through P and Q
- (1) c. Find the size of θ , correct to the nearest degree
- (3) d. Find the equation of the QR, and hence find the co-ordinates of R
- (2) e. Find the perpendicular distance from P to QR

2003

Question 3

S/A
T/C

YEAR 11 TERM 3

COMMON TEST
2 UNIT

Question 1

a) $\tan 30^\circ = 1/\sqrt{3}$ ①

b) $x^3 - 27 = (x-3)(x^2 + 3x + 9)$ ①

$$\begin{aligned} \text{ii. } x^2 - xy - 3x + 3y \\ = x(x-y) - 3(x-y) \quad ① \\ = (x-y)(x-3) \quad ① \end{aligned}$$

$$\begin{aligned} \text{c) } |2x+1| \leq 7 \\ 2x+1 \leq 7 \quad -2x-1 \leq 7 \\ 2x \leq 6 \quad -2x \leq 8 \\ x \leq 3 \quad ① \quad x \geq -4 \quad ① \end{aligned}$$

$$\begin{aligned} \text{d) } 3x^2 - 14x - 5 = 0 \\ 3x+1 \quad x-5 \\ (3x+1)(x-5) = 0 \\ x = -\frac{1}{3} \quad ①, \quad x = 5 \quad ① \end{aligned}$$

Question 2

$$\begin{aligned} \text{a) } \frac{3}{\sqrt{7}+2} \times \frac{\sqrt{7}-2}{\sqrt{7}-2} &= \frac{3(\sqrt{7}-2)}{7-4} \\ &= \sqrt{7}-2 \quad ① \end{aligned}$$

$$\begin{aligned} \text{b) } 2x + 3y &= -1 \\ 9x - 3y &= 12 \quad ① \end{aligned}$$

$$\begin{aligned} 11x &= 11 \\ x &= 1 \quad ① \end{aligned}$$

$$y = -1 \quad ①$$

$$\text{c) } \angle DBC = 60^\circ \quad (\triangle BDC \text{ is equilateral}) \quad ①$$

$$\angle ABD = 85^\circ \quad (\text{opp } \angle \text{'s in } \triangle \text{ equal})$$

$$\therefore \angle ABC = 60 + 85 = 145^\circ \quad (\text{sum of adj } \angle \text{'s})$$

$$\begin{aligned} \text{a) i. } \cos 225^\circ &= -\cos 45^\circ \\ &= -1/\sqrt{2} \quad // \end{aligned}$$

$$\text{ii. } \sec(-30^\circ) = \frac{1}{\cos 30^\circ} = \frac{2}{\sqrt{3}} \quad //$$

$$\begin{aligned} \text{iii. } \tan^2 390^\circ &= (\tan 30^\circ)^2 \\ &= (1/\sqrt{3})^2 \quad \checkmark \\ &= 1/3 \end{aligned}$$

$$\begin{aligned} \text{b) } \cos(360-\theta) \sin(180+\alpha) \\ \text{(4th quad)} \quad \text{(3rd quad)} \\ = \cos \theta - \sin \alpha \quad \checkmark \\ = p - q \\ = -pq \quad \checkmark \end{aligned}$$

Question 4

$$\begin{aligned} \text{a) } \sin 2x &= \cos(40+x) \\ 2x + 40 + x &= 90 \\ 3x &= 50 \\ x &= 50/3 \quad \checkmark \quad ① \end{aligned}$$

$$\begin{aligned} \text{b) i. } \frac{x}{\sin 68^\circ} &= \frac{15.6}{\sin 78^\circ} \quad ① = 78^\circ \\ x &= \frac{15.6 \sin 68^\circ}{\sin 78^\circ} \quad ① = \text{Sine rule} \\ &= 14.79 \text{ m} \quad ① \text{ Ans correct to 2 dec pl.} \end{aligned}$$

$$\begin{aligned} \text{ii. } A &= \frac{1}{2} ab \sin C \quad ① \\ \text{use ans to part (i)} \quad &= \frac{1}{2} \times 15.6 \times 14.79 \times \sin 34^\circ \\ &= 64.509... \\ &= 65 \text{ m}^2 \quad > ① \text{ either} \end{aligned}$$

$$\text{c) } (x, y) \quad (7, 2) \quad \text{Midpt} \quad (1, -3)$$

$$\frac{x+7}{2} = 1 \quad \frac{y+2}{2} = -3$$

$$x+7=2$$

$$x = -5 \quad ①$$

$$y+2=-6$$

$$y = -8 \quad ①$$

$$\therefore B(-5, -8)$$

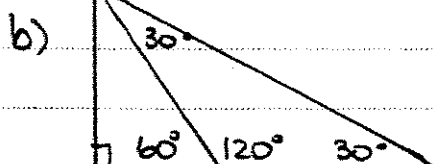
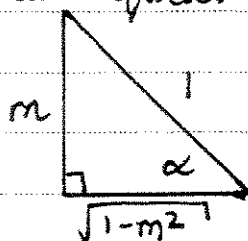
Question 5

2nd quad

a) $\sin \alpha = m$

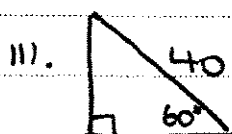
$\therefore \cos \alpha = -\sqrt{1-m^2}$

$= -\sqrt{1-m^2}$ ①



i. $\angle ACD = 120^\circ$ ①

ii. $AC = 40$ m $\triangle ACD$ is ①
Isosceles.



$\sin 60^\circ = \frac{AB}{40}$ ①

$AB = 40 \sin 60^\circ$

$= 40 \frac{\sqrt{3}}{2}$ ①

$= 20\sqrt{3}$ ①

Question 6

a) $\cos x = -\frac{1}{2}$ $-180^\circ \leq x \leq 180^\circ$

$x = 120^\circ, -120^\circ$

①

①

$$\begin{array}{c} \textcircled{S} | A \\ \textcircled{T} | C \\ \hline \cos 60^\circ = \frac{1}{2} \end{array}$$

b) $3\tan^2 x - 1 = 0$

$\tan x = \pm \frac{1}{\sqrt{3}}$ ①

$x = 30^\circ, 150^\circ, 210^\circ, 330^\circ$ ②

c) $\sec 2x = 1$

$\therefore \cos 2x = 1$ ①

$2x = 0^\circ, 360^\circ, 720^\circ$ ①

$x = 0^\circ, 180^\circ, 360^\circ$ ①

Question 7

a) $LHS = \frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta}$

① $= \frac{\sin^2 \theta + (1 + \cos \theta)^2}{\sin \theta (1 + \cos \theta)}$

① $= \frac{\sin^2 \theta + 1 + 2\cos \theta + \cos^2 \theta}{\sin \theta (1 + \cos \theta)}$

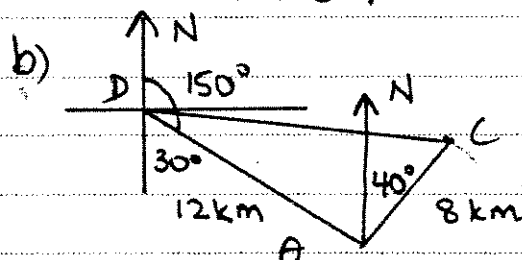
$= \frac{1 + 1 + 2\cos \theta}{\sin \theta (1 + \cos \theta)}$

$= \frac{2(1 + \cos \theta)}{\sin \theta (1 + \cos \theta)}$

① $= \frac{2}{\sin \theta}$

$= 2 \operatorname{cosec} \theta$

$= RHS$



i. $\angle DAC = 30^\circ + 40^\circ$
 $= 70^\circ$

ii. $x^2 = 12^2 + 8^2 - 2 \times 12 \times 8 \times \cos 70^\circ$
 $x = 11.93$ km (2dp)

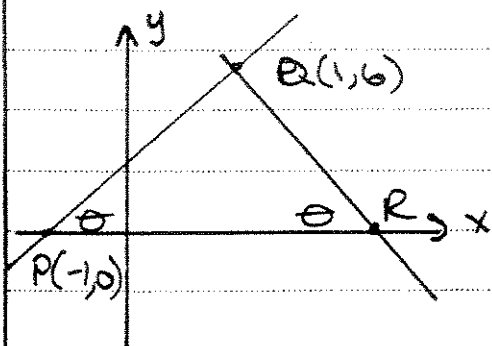
$$\begin{array}{c} 11.93 \\ \triangle \\ 12 \quad 8 \end{array} \quad \frac{\sin \alpha}{8} = \frac{\sin 70^\circ}{11.93}$$

$$\sin \alpha = \frac{\sin 70^\circ \times 8}{11.93}$$

$\alpha = 39^\circ$

$\therefore \text{bearing} = 150 - 39$
 $= 111^\circ T$

Question 8



$$\begin{aligned}
 \text{a) } d &= \sqrt{(1 - (-1))^2 + (6 - 0)^2} \\
 &= \sqrt{2^2 + 6^2} \\
 &= \sqrt{40} \quad \text{--- ①} \\
 &= 2\sqrt{10}
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } m_{PQ} &= \frac{6 - 0}{1 - (-1)} \\
 &= \frac{6}{2} \\
 &= 3 \quad \text{①}
 \end{aligned}$$

$$\begin{aligned}
 \therefore y - 6 &= 3(x - 1) \\
 y &= 3x + 3 \quad \text{①}
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } \tan \theta &= m \\
 \tan \theta &= 3 \\
 \theta &= 72^\circ \quad \text{①}
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } m_{QR} &= -3 \quad \text{① as } \angle QRP = \theta \\
 \therefore y - 6 &= -3(x - 1) \\
 y - 6 &= -3x + 3 \\
 3x + y - 9 &= 0 \quad \text{①}
 \end{aligned}$$

$$\begin{aligned}
 \therefore R \text{ when } y &= 0 \\
 x &= 3
 \end{aligned}$$

$$R(3, 0) \quad \text{①}$$

$$\begin{aligned}
 \text{e) } d_{\perp} &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} > \text{either } \text{①} \\
 &= \frac{|3x + y - 9|}{\sqrt{9 + 1^2}} \text{ pt } (-1, 0) \\
 &= \frac{|-3 + 0 - 9|}{\sqrt{10}} \\
 &= \frac{12}{\sqrt{10}} \quad \text{①} \\
 &= \frac{6\sqrt{10}}{5}
 \end{aligned}$$